

Electrostatic features of PL25 M23 compared with reference SLT/M23 and M23 enzymes

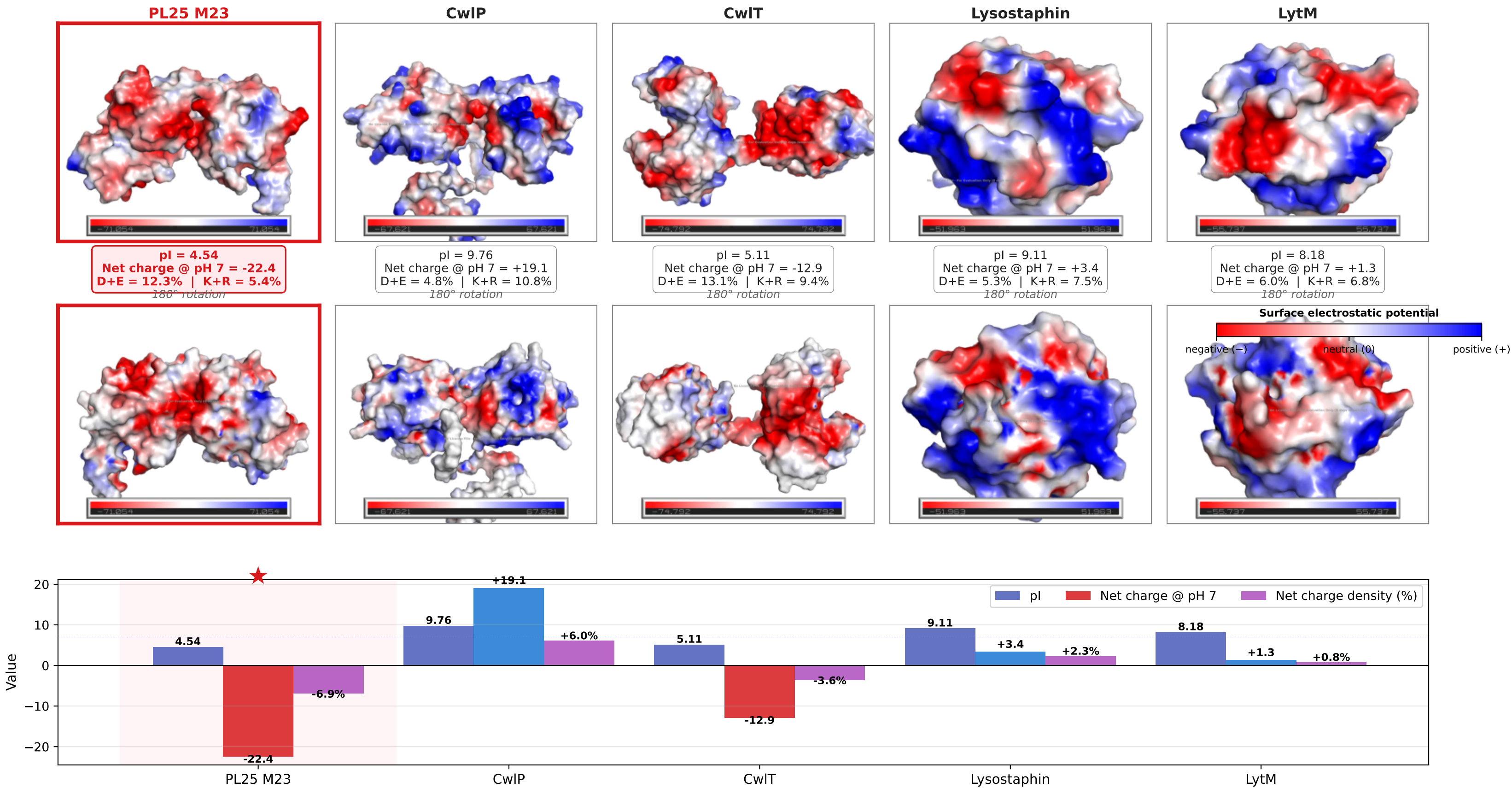


Figure 8. Electrostatic comparison of PL25 M23 and reference cell-wall hydrolases. Surface electrostatic potentials are shown for PL25 M23, CwIP, CwIT, lysostaphin, and LytM, with red indicating negative potential, blue positive potential, and white neutral regions. PL25 M23 (red outline, bold) showed the lowest predicted pI (4.54), strongest negative net charge at pH 7 (−22.4), and high acidic residue content (D+E = 12.3 %), consistent with an acidic surface profile associated with halophilic protein adaptation. Values were calculated for the modeled catalytic/mature regions used for structural comparison.